

PROPOSAL V1 - SHALE GAS SHOCK

Wooyong Park
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Research Questions

- How did shale gas shock affect education, income, and other living standards (was there a dutch disease)

Methodologies

Notes

- For education, I included age 18-25 individuals to the sample
- For income, I included age 18-40 individuals to the sample

Synthetic Control

- Control variables for donor pool: `predictors = c("white_ratio", "black_ratio", "asian_ratio", "hispanic_ratio", "youth_ratio", "population")`

Trial (1)

First, I tried SCM with Texas as the treatment group. Without consideration of other shale-gas states like Oklahoma or Penn, I had the following results.

Education

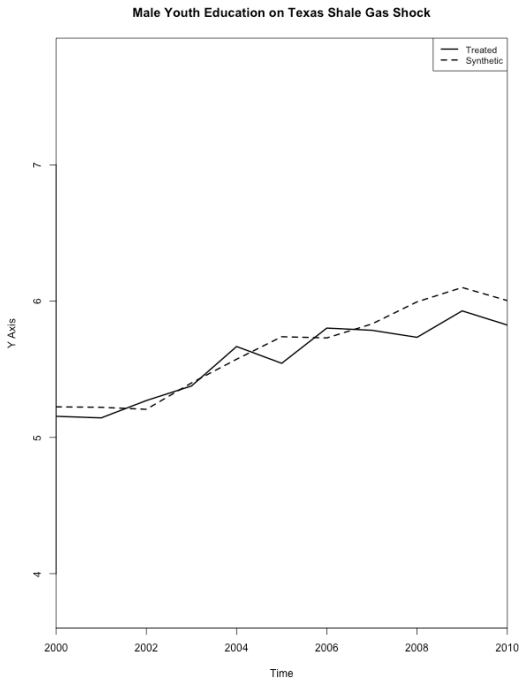
- In both genders, we find a slight decrease in the avg. years of education, but in a very modest degree.

Income

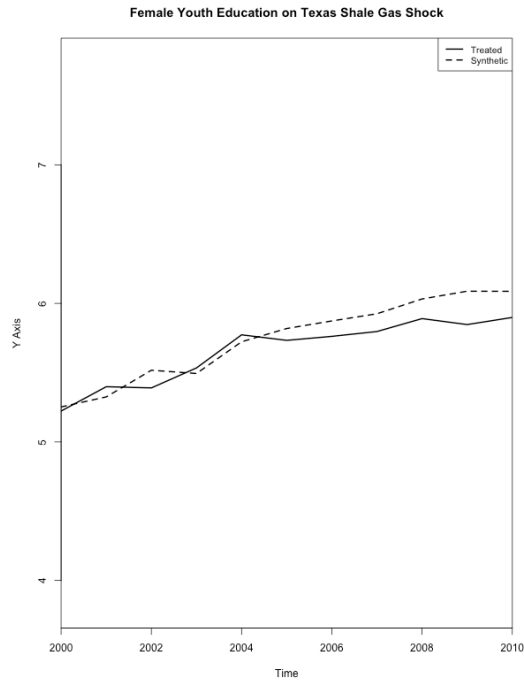
- In both genders, we find a slight increase in the avg. income, but in a very modest degree.

Problems with Trial (1)

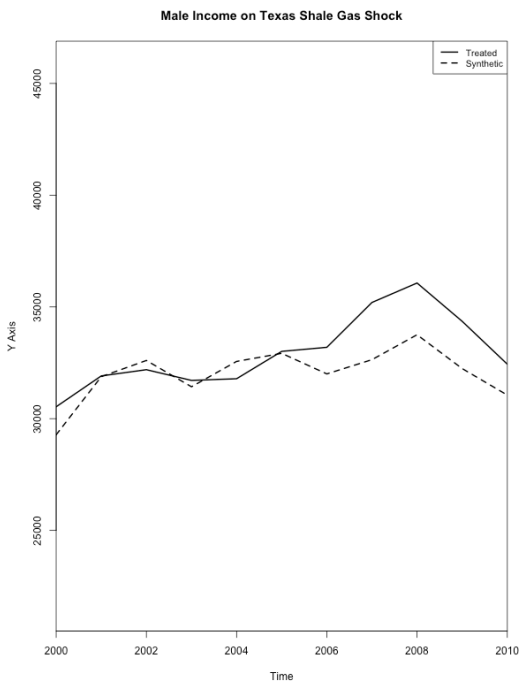
1. Placebo test results are so bad. For example, in education,
 - Note: A good MSPE test result should have the treated (gray) column on the far right.
2. I didn't account for other states with shale gas activity.



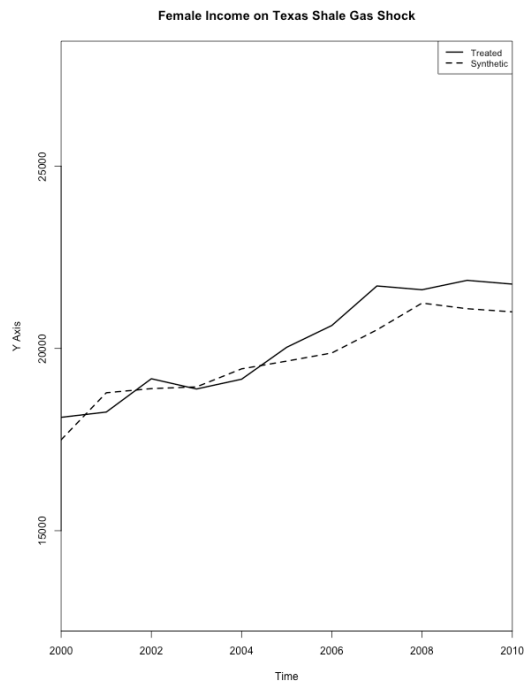
(a) scm-male-edu



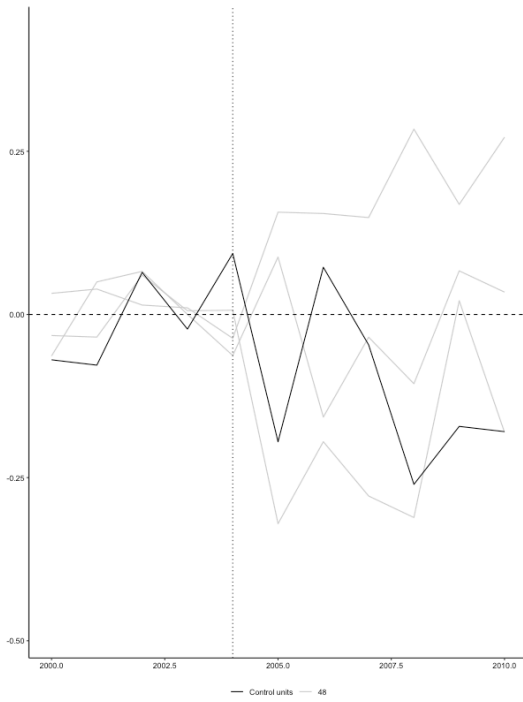
(a) scm-female-edu



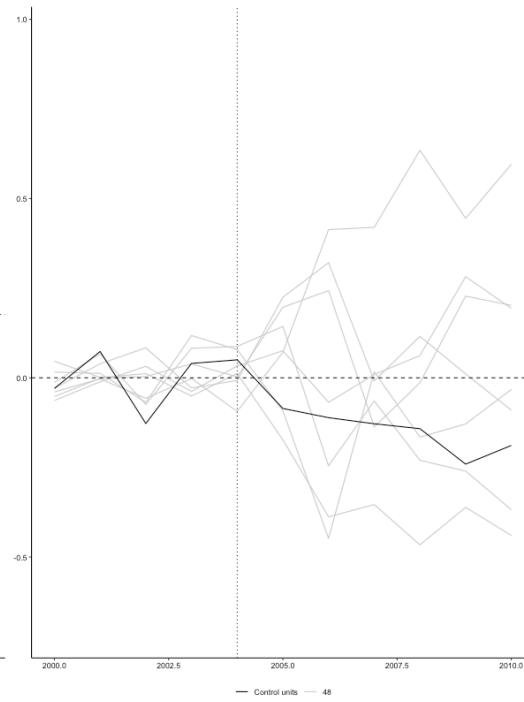
(a) scm-male-income



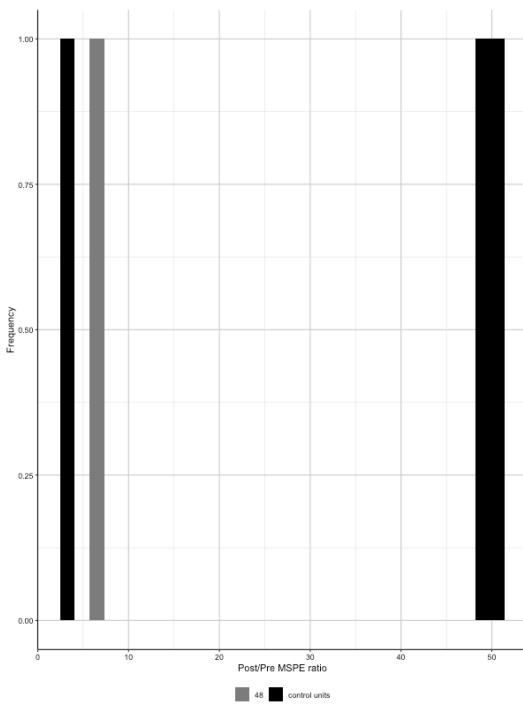
(a) scm-female-income



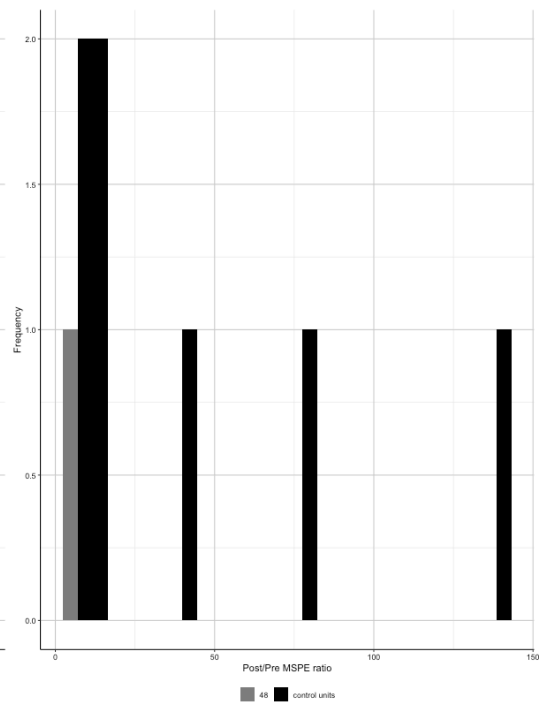
(a) scm-male-edu-placebo



(a) scm-female-edu-placebo



(a) scm-male-edu-mspe



(a) scm-female-edu-mspe

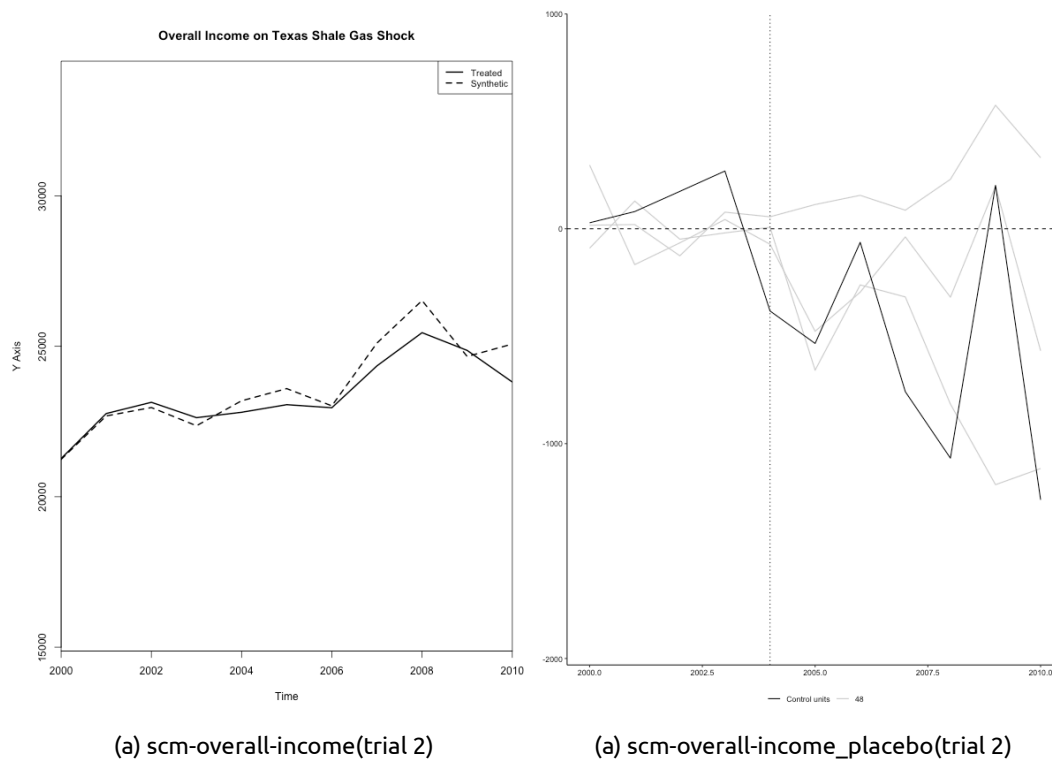
Trial (2)

In trial (2), I excluded other states with shale-gas activity.

Results are unsatisfactory:

- Even the slightest amount of different trends between SC and treated disappeared
- Placebo tests are not working well.

For example, the average income effect is as follows:



DID

Although I did not have a straight DID now, I made plots equivalent to DID analyses.

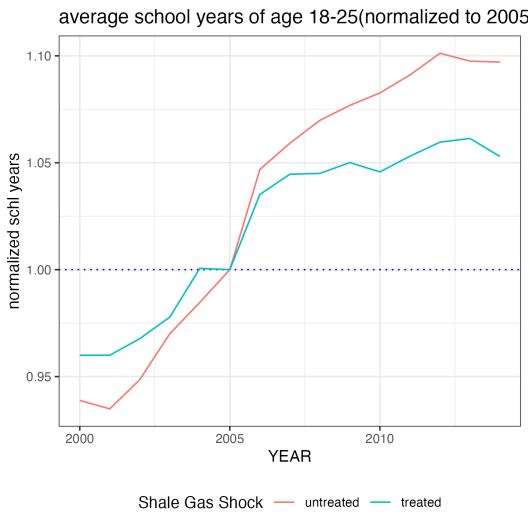
This looks a bit more satisfying:

Overall Effect

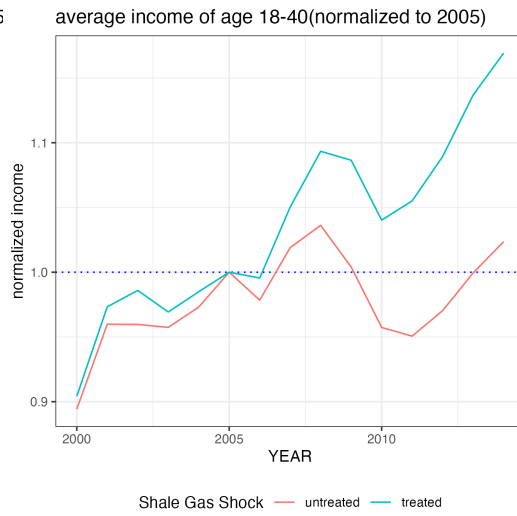
By sex

To-dos

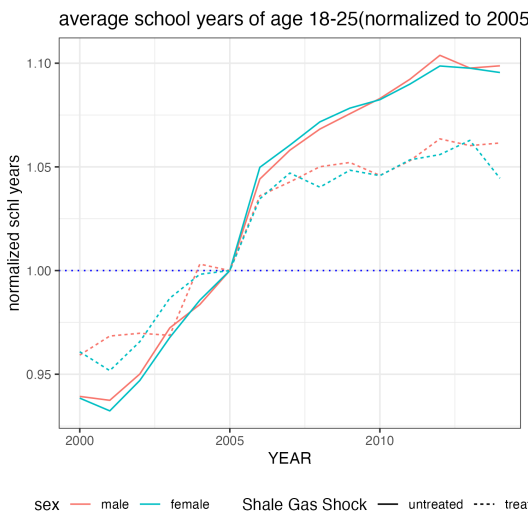
- Formally rerun DID
- Search for any individual-level panel data with **state, year(2000-2010), sex, age** information. It'd be better if it includes **education, income, labor market participation**.
- Extend our question:



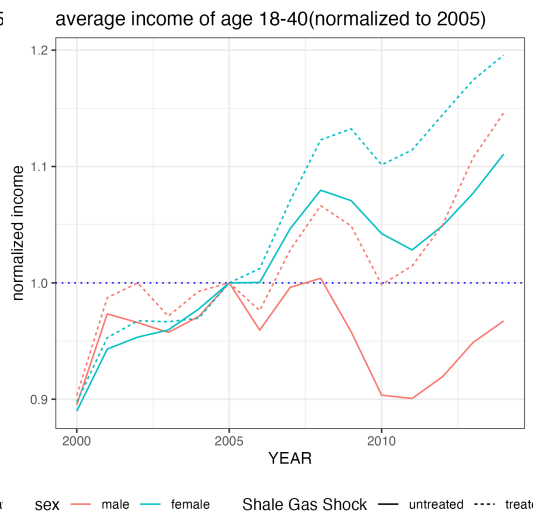
(a) did-schlyears



(a) did-income



(a) did-schlyears-by_sex



(a) did-income-by_sex

- Heterogeneous Effects on men and women? Why? For example, male income decreased in nonshale states.
- What are the channels of the income/education effects?
- Why did SCM do bad but DID do great? Econometric and causal inference-based answers

Code Directory

- Synthetic control:
 - trial (1) : final-project/code/scm.r
 - trial (2) : final-project/code/scm-2.R
- DID: final-project/code/did-with-panel.r